1 Numerical Setup

This section describes the numerical methods and model configuration used to simulate the double-explosion scenario consisting of a Type IIb supernova (SN_A) and a subsequent H/He shell ejection into the SN_A ejecta (SN_B). The modelling pipeline couples the multi-group radiation hydrodynamics code STELLA (Blinnikov et al., 1998, 2006; Blinnikov & Sorokina, 2000) with the non-LTE radiative transfer code CMFGEN (Dessart et al., 2015; Hillier & Dessart, 2012; Hillier & Miller, 1998) for spectral synthesis at selected epochs.

1.1 Overview of the Modelling Pipeline

The simulation proceeds in two stages. In the first stage (SN_A), a stripped-envelope progenitor is exploded as a Type IIb supernova and evolved with STELLA from shock breakout (SBO) through the radioactive-decay-powered light-curve phase. A snapshot at $\sim$30 d post-SBO is extracted and remapped onto a CMFGEN depth grid for non-LTE spectral synthesis. The SN_A ejecta is then homologously evolved to late times ($\sim$400 d) to serve as the circumstellar medium (CSM) for the second stage.

In the second stage (SN_B), a H/He shell is placed at the centre of the homologously expanded SN_A ejecta and launched at 5000 km s^{-1} into the surrounding CSM. STELLA evolves the shell–CSM interaction, and a snapshot at $\sim$20–40 d after shell launch is extracted for CMFGEN spectral synthesis.

The entire pipeline is orchestrated by a single workflow manager (`run_model_grid.py`) that handles progenitor conversion, STELLA job submission, snapshot extraction, CMFGEN grid generation, and dependency-chained spectral synthesis jobs on a high-performance computing (HPC) cluster.

1.2 SN_A: Type IIb Explosion of a Stripped Helium Star

1.2.1 Progenitor Model and Pre-SN Conversion

The SN_A progenitor is based on the solar-metallicity 12 $M_\odot$ model `s12.0` from Woosley & Heger (2007), evolved through core collapse with the KEPLER stellar evolution code. The hydrogen envelope is removed to mimic binary stripping, leaving a $\sim$3.0 $M_\odot$ helium core. This configuration matches model 12C of Ergon et al. (2022), which was constructed to reproduce the Type IIb SN 2011dh (Ergon et al., 2015).

The KEPLER pre-SN model is converted to STELLA's Lagrangian hydrodynamics format by the script `convert_kepler_presn_to_stella.py`, which performs the following operations:

1. **Mass cut excision:** The innermost 1.4 $M_\odot$ (the compact remnant) is removed, leaving $\sim$1.6 $M_\odot$ of ejecta.

2. **Composition remapping:** KEPLER isotopes are mapped onto STELLA's 15-element nuclear network (H, He, C, N, O, Ne, Na, Mg, Al, Si, S, Ar, Ca, Fe, Ni). The iron-group isotopes (^{56}Fe, ^{56}Ni, ^{44}Ti, ^{40}Ca, ^{36}Ar, ^{32}S) are combined into a single Fe-peak tracer.

3. **H surface layer addition:** A thin hydrogen envelope of 0.03–0.10 $M_\odot$ with H mass fraction $X = 0.7$ is added to the outermost zones, reproducing the residual H layer characteristic of Type IIb progenitors (Ergon et al., 2022).

4. **CSM wind attachment:** A low-density $1/r^2$ circumstellar wind is appended beyond the stellar surface using a sigmoid-blended transition that ensures C^∞ continuity in both the density and all its derivatives. The wind has a mass-loss rate $\dot{M} = 10^{-6}\,M_\odot\,\mathrm{yr}^{-1}$, a terminal velocity $v_{\rm wind} = 100\,\mathrm{km\,s^{-1}}$, and a duration of 320 yr, yielding an outer radius of $\sim 10^{17}$ cm and a total CSM mass of $\sim 3 \times 10^{-4}\,M_\odot$. The CSM is primarily required for numerical stability — it prevents the STELLA radiation solver from stalling when the photosphere exits the compact ejecta at $\sim$10 d — while contributing negligible mass and optical depth to the light curve.

5. **Density gradient smoothing:** Steep density gradients ($|d\log\rho/d\log r| > 5$) in the pre-SN profile are smoothed with a Gaussian kernel ($\sigma = 3$ zones) to prevent GEAR solver stalls at the initial condition.

The resulting profile is remapped onto 500 Lagrangian zones (450 ejecta zones + 50 wind zones) and written in STELLA's `.hyd` (hydrodynamics) and `.abn` (abundance) binary formats.

1.2.2 STELLA Radiation Hydrodynamics

STELLA (Blinnikov et al., 1998, 2006; Blinnikov & Sorokina, 2000) solves the one-dimensional Lagrangian radiation hydrodynamics equations with multi-group grey radiative transfer in the homologous comoving frame. The code employs a variable Eddington factor closure and treats radiation transport in two regimes: a diffusion approximation for the optically thick interior ($\tau >$ `FitTau`) and a ray-by-ray (characteristic) method for the optically thin outer layers ($\tau <$ `FitTau`). The boundary between these regimes, $N_{\rm cnd}$, is determined at each timestep by searching for the depth where the optical depth at the minimum-opacity frequency equals `FitTau`.

Explosion mechanism. The explosion is driven by a thermal bomb deposited in the inner 0.3 $M_\odot$ of the ejecta over a duration of 10 s, with a total energy input of 0.9×10^{51} erg. The shorter bomb duration (10 s versus the 100 s used in earlier runs) increases the kinetic-energy conversion efficiency

from ∼61% to ∼80%, yielding a final kinetic energy of ∼0.7 foe consistent with the Ergon et al. (2022) value for SN 2011dh. The parameter $E_{\rm KO} = 0$ is set so that STELLA preserves the zero initial velocities from the `.hyd` file; the thermal bomb generates the explosion and establishes the velocity profile during the first ∼0.1 d of evolution.

Radioactive heating. A ^{56}Ni mass of $0.075\,M_\odot$ (Ergon et al., 2022) is distributed uniformly over the inner $0.3\,M_\odot$ of the ejecta (the inner 40% of zones). The ^{56}Ni is read from a dedicated `.xni` file and injected into the composition via the `eve2` pre-processor. The radioactive decay chain $^{56}\mathrm{Ni} \rightarrow^{56}\mathrm{Co} \rightarrow^{56}\mathrm{Fe}$ is followed self-consistently by STELLA's nuclear network, with gamma-ray deposition treated using the Swartz et al. (1995) opacity formalism.

Numerical parameters. The key STELLA control parameters for the SN_A model are summarised in Table 1. The GEAR integrator (Byrne & Hindmarsh, 1975; Gear, 1971) is used with a relative tolerance of $\epsilon = 3 \times 10^{-3}$, a minimum timestep of 10^{-14} s, and a maximum timestep of ∼5.8 d. The artificial viscosity parameters ($A_Q = 50$, $B_Q = 1$) are set high to damp the thermal-bomb shock at early times ($t < 1$ d); lower values ($A_Q = 20$) caused matrix solver failures at $t \sim 7 \times 10^{-5}$ d.

The `FitTau` parameter is set to 5.0, following the STELLA reference model `X_scat2.dat`. This value is appropriate for SN ejecta with strongly frequency-dependent opacity, where the optical depth at the minimum-opacity frequency is ∼100× smaller than the grey optical depth. Setting `FitTau` = 15 (as in the homogeneous test model `m030307mhomo.dat`) causes the $N_{\rm cnd}$ boundary to collapse prematurely when the minimum-frequency τ drops below 15, leading to a catastrophic jump in the diffusion–ray-by-ray boundary and a failure of the Y12M sparse matrix solver.

To prevent $N_{\rm cnd}$ collapse at late times ($t > 10$ d), where the minimum-frequency optical depth inevitably drops below any reasonable `FitTau` due to geometric expansion, the minimum number of diffusion zones is set to $N_{\rm CNMIN} = N_{\rm zon}/10 = 50$. Additionally, the per-timestep change in $N_{\rm cnd}$ is limited to $\Delta N_{\rm cnd} = 20$ zones, converting catastrophic single-step jumps (e.g. 279→50) into a gradual descent over ∼12 timesteps that the matrix solver can tolerate.

1.2.3 Homologous Evolution to Late Times

At late times ($t \gtrsim 60$ d), the SN_A ejecta is in free expansion ($v = r/t$), and the evolution is accurately described by homologous scaling. The script `homologous_evolution.py` scales the final STELLA snapshot to a target

Table 1: STELLA numerical parameters for the SN_A and SN_B models.

Parameter	SN_A	SN_B
Number of zones ($N_{\rm zon}$)	500	500
Target duration (d)	100	45
Explosion energy (10^{51} erg)	0.9 (thermal bomb)	0 (kinetic)
$E_{\rm KO}$	0	0
A_Q (artificial viscosity)	50	10
B_Q (pseudo-viscosity)	1	1
`FitTau`	5.0	5.0
$N_{\rm CNMIN}$	$N_{\rm zon}/10$	$N_{\rm zon}/10$
$\Delta N_{\rm cnd}$ limit	20	20
GEAR method	2	3
ϵ (integrator tolerance)	3×10^{-3}	3×10^{-3}
$h_{\rm min}$ (s)	10^{-14}	10^{-14}
$h_{\rm max}$ (s)	5×10^5	3600
$T_{\rm floor}$ (K)	300	300
$T_{\rm rad,\,floor}$ (K)	100	100
^{56}Ni mass ($M_\odot$)	0.075	0
Scattering	Enabled	Enabled

age $t_{\rm target}$ using:

$$r(t) = r(t_0)\,\frac{t}{t_0}, \tag{1}$$

$$v(t) = v(t_0) \quad \text{(constant)}, \tag{2}$$

$$\rho(t) = \rho(t_0)\left(\frac{t_0}{t}\right)^3, \tag{3}$$

$$T(t) = T(t_0)\,\frac{t_0}{t} \quad \text{(adiabatic)}, \tag{4}$$

where t_0 is the age of the last STELLA epoch. These relations are valid when the expansion is highly supersonic, there is no ongoing energy injection (the ^{56}Ni decay chain is negligible at $t \gtrsim 200$ d), and there is no significant CSM interaction. The homologously evolved profile at 400 d serves as the CSM for the SN_B model.

1.3 SN_B: H/He Shell Interaction with the SN_A Ejecta

1.3.1 Initial Model Construction

The SN_B initial model is constructed by the script `prepare_sn_b.py` (or `prepare_x_model.py` for the three-component variant with a connecting wind). The model consists of two (or three) components, assembled from inside out:

1. **H/He shell (inner):** A uniform-density sphere of $1.0\,M_\odot$ with composition $X = 0.7$, $Y = 0.3$ and a homologous velocity profile reaching $5000\,\mathrm{km\,s^{-1}}$ at its outer edge. The shell represents material ejected by a secondary explosion or a pulsational mass-loss event. The density structure follows a power-law $\rho \propto r^{-n_\rho}$ with $n_\rho = 8$, consistent with the reverse-shock shell models of Dessart et al. (2016) for SN 1994W. The temperature profile follows $\rho \propto r^{-n_T}$ with $n_T = 0.4$, with a maximum core temperature of 65 000 K.

2. **Connecting wind (middle, optional):** A $1/r^2$ wind with $\dot{M} = 0.1\,M_\odot\,\mathrm{yr^{-1}}$ and $v_\mathrm{wind} = 100\,\mathrm{km\,s^{-1}}$ bridges the shell and the old ejecta, preventing a hard density discontinuity that would stall the GEAR solver. This component is included in the three-component X model variant.

3. **SN_A ejecta (outer):** The 400-day homologously evolved SN_A profile (§1.2.3), which now acts as the CSM. The ejecta retains its homologous velocity structure and He-rich, metal-enhanced composition from the Type IIb explosion.

The components are combined into a single 500-zone STELLA `.hyd` file. The parameter $E_\mathrm{KO} = 0$ is essential here: it preserves both the shell velocity ($5000\,\mathrm{km\,s^{-1}}$) and the CSM homologous velocity profile. Setting $E_\mathrm{KO} > 0$ would overwrite all velocities with a triangular (linear-in-mass) profile, destroying the physically motivated shell–CSM velocity contrast and causing the solver to stall.

1.3.2 STELLA Evolution of the Interaction

STELLA evolves the SN_B model for 45 d from the shell launch time. The H/He shell, moving at $5000\,\mathrm{km\,s^{-1}}$, drives a strong forward shock into the slower CSM ($v \sim 500$–$6000\,\mathrm{km\,s^{-1}}$), while a reverse shock propagates back into the shell. The shock interaction converts kinetic energy into radiation, producing the interaction-powered luminosity that characterises Type IIn supernovae (Chugai & Danziger, 2001; Dessart et al., 2016).

The artificial viscosity is set to $A_Q = 10$ (higher than the SN_A value of 2–4 for a standard thermal-bomb explosion) to stabilise the GEAR solver across the strong velocity reversal at the shell–CSM interface. The GEAR method is set to 3 (GEAR+BGH) for improved stability in the stiff interaction system. No ^{56}Ni is included in the SN_B model; the luminosity is entirely interaction-powered. The maximum timestep is limited to 3600 s (1 h) to resolve the early shock interaction phase.

1.4 CMFGEN Non-LTE Spectral Synthesis

1.4.1 STELLA-to-CMFGEN Conversion

At selected epochs (30 d post-SBO for SN_A; 20–40 d post-launch for SN_B), a snapshot is extracted from the STELLA `.swd` output file and converted to CMFGEN's `SN_HYDRO_DATA` format by the script `convert_stella_swd_to_cmfgen.py`. The conversion performs the following steps:

1. **Epoch selection:** The STELLA epoch closest to the requested snapshot day (within a tolerance of ± 2 d) is selected. If no STELLA epoch is available (e.g. for late-time snapshots beyond the STELLA target), the homologous evolution procedure (§1.2.3) is applied.

2. **Optical-depth filtering:** Zones are filtered to retain only the region $\tau_{\rm min} \leq \tau \leq \tau_{\rm max}$ (typically $0.01 \leq \tau \leq 30$), where τ is computed from the STELLA NLTE opacity. This removes both the optically thin vacuum exterior and the optically thick interior where CMFGEN's Sobolev approximation breaks down.

3. **Adaptive grid remapping:** The filtered STELLA zones are remapped onto a CMFGEN depth grid using an adaptive algorithm that concentrates zones in regions of steep gradients (ionisation fronts, density jumps, velocity reversals). The target grid size is $N_D \sim 150$–200 points, with constraints on maximum cell-to-cell jumps in radius ($\Delta r/r \leq 0.05$), temperature ($\Delta \log T \leq 0.10$), density ($\Delta \log \rho \leq 0.15$), and velocity ($\Delta v \leq 0.30\, v$). The velocity–density structure parameter $\sigma = d\ln v / d\ln r - 1$ is monitored, and additional zones are inserted where $|\sigma|$ or $|d\sigma/dr|$ exceeds threshold values, ensuring adequate resolution of the Sobolev line optical depth.

4. **Composition transfer:** The STELLA 15-element composition is mapped onto CMFGEN's species list. The ^{56}Ni distribution is read from the STELLA `.xni` file (Fortran unformatted REAL*8) and included in the `SN_HYDRO_DATA` abundance table so that CMFGEN can compute radioactive decay heating self-consistently.

5. **Velocity smoothing:** Small numerical oscillations in the STELLA velocity profile ($\pm 0.3\,\mathrm{km\,s^{-1}}$) are smoothed with a Savitzky–Golay filter (window = 5 zones, $\Delta v = 50\,\mathrm{km\,s^{-1}}$) to prevent the structure parameter σ from reaching unphysical values ($\sigma < -19$) that destabilise the CMFGEN Sobolev integrals.

The output `SN_HYDRO_DATA` file is written in CMFGEN's outer-to-inner (decreasing radius) depth convention, with radius in units of 10^{10} cm, velocity in $10^5\,\mathrm{cm\,s^{-1}}$, temperature in 10^4 K, and density in $\mathrm{g\,cm^{-3}}$.

1.4.2 CMFGEN Configuration and Convergence Strategy

CMFGEN (Dessart et al., 2015; Hillier & Dessart, 2012; Hillier & Miller, 1998) computes non-LTE level populations and emergent spectra for a spherically symmetric supernova atmosphere. The code solves the radiative transfer equation in the Sobolev approximation with continuum opacities treated in the co-moving frame, and iterates the statistical equilibrium equations coupled to the radiative transfer via an accelerated Lambda iteration (ALI) scheme (Rybicki & Hummer, 1991).

Velocity law and grid. For both SN_A and SN_B, the velocity law is set to `VEL_LAW = 12`, which reads the actual (non-homologous) velocity profile from the `SN_HYDRO_DATA` input rather than fitting a homologous $v \propto r$ relation. This is essential for the SN_B model, where the shell–CSM velocity reversal produces $\sigma < -1$ regions that a homologous fit cannot represent. The `VEL_LAW = 12` routine includes a protective clamp that sets $\sigma = -0.999$ when $-1.2 < \sigma < -1.0$, preventing the Sobolev line optical depth denominator $|1 + \sigma\mu^2|$ from approaching zero. Empirical tests confirm that removing this clamp causes catastrophic divergence of the Ng acceleration extrapolation, even though the Dessart et al. (2015) Sobolev guard $\max(|1 + \sigma\mu^2|, -0.2(1 + \sigma), 0.1)$ is algebraically well-defined for all finite σ.

Solver configuration. The depth-coupling solver is set to `TRIDIAG` (`NUM_BNDS = 3`), which allows inter-depth error propagation via a tridiagonal band matrix. This is required for models with non-monotonic velocity ($\sigma < -1$), where the diagonal-only (`DIAG`) solver cannot capture the inter-depth coupling introduced by the velocity reversal. The formal solution uses `N_TYPE = N_ON_J` with `USE_LAM_ES = T`, which sets the mean intensity J to the formal-solution value $J_{\rm TC}$ and bypasses the moment equations, improving stability for models with large velocity gradients.

Iteration strategy. The convergence strategy employs alternating LAMBDA and Boltzmann Acceleration (BA) iterations, controlled by the following VADAT parameters:

- `DO_NG = F`: Ng acceleration (Ng, 1974) is disabled. While Ng acceleration can speed convergence for well-behaved models, it destabilises trace species (e.g. Fe IV, Fe V) at ionisation fronts by extrapolating across oscillatory population changes, leading to divergence of up to 10^{26}% in the maximum correction.
- `DO_AV = T`: Population averaging (`AVE_FLIPS`) is enabled to damp oscillating populations between LAMBDA iterations. A source-code patch was applied to allow `AVE_FLIPS` to fire during continuous LAMBDA iterations (`DO_LAM_IT = T`), which was previously blocked by a condition

that required `MAIN_COUNTER > LAST_LAMBDA + NUM_OSC_AV + 2` — a condition that is never satisfied when every iteration is a LAMBDA iteration.

- `MAX_LAM = 10^5`: The maximum fractional population change per LAMBDA iteration is capped at 10^5, with `LAM_SCALE_OPT = LIMIT` applying a proportional limiter to trace species.
- `MAX_LIN = 3`: The maximum number of linearisation steps per BA iteration is limited to 3 (reduced from the manual default of 10) for cold-start stability, as the BA solve can produce extreme corrections when the initial LTE populations are far from the converged solution.
- `BA_CHK_FAC = 0.05`: The BA matrix is held fixed when the maximum population correction falls below 5%, avoiding unnecessary matrix recomputations.
- `METHOD = ZERO`: The trapezoidal optical-depth integration method is used, as recommended by the CMFGEN documentation for models with convergence difficulties.
- `FIX_T = T`: The temperature is fixed to the hydrodynamic (STELLA) value at all depths throughout the iteration, preventing CMFGEN from solving for temperature and using the physically consistent hydrodynamic temperature instead.

Inner boundary condition. The inner boundary condition is set to `DIFFUSION` (`DIF = T`, `IB_METH = DIFFUSION`) when the inner optical depth $\tau_{\rm inner} \geq 50$, providing a physically motivated radiation field seed ($L_\star$) from the STELLA luminosity. When $\tau_{\rm inner} < 50$, the `DIFFUSION` boundary is still used (with a warning), as a non-zero $L_\star$ is preferable to the `ZERO_FLUX` alternative, which provides no radiation seeding and produces $L = 0$ at iteration 1.

Species selection. The atomic model includes H I, He I, O I, Si I–II, Ca I–II, and Fe I–III as the primary spectral-line species. High-ionisation iron species (Fe IV, Fe V) are removed for the 30 d SN_A snapshot, as their ionisation potentials ($\chi = 54.8$ and 77 eV) are far above the maximum ejecta temperature ($T_{\rm max} \approx 7700$ K at 30 d), making their NLTE populations $\sim 10^{-26}$ (set by photoionisation from the photospheric radiation field) and introducing 10^{16}% trace-species noise into the convergence. Similarly, C II is reduced to ground-state-only (`ISF = 1,1,1`) for models with near-zero carbon abundance ($X_C \sim 10^{-15}$), as 8 NLTE levels with zero abundance drove 10^{36} fractional population cascades.

Spectral synthesis. Once the level populations have converged, CMFGEN computes the emergent spectrum via a formal solution of the radiative transfer equation on a fine frequency grid spanning 900–28 000 Å (covering the optical through mid-infrared for JWST MIRI comparison). The spectrum includes line blanketing from all included species, continuum opacities (bound-free, free-free, electron scattering), and the diffuse emission from the SN ejecta. An observed-frame spectrum is produced by applying the cosmological redshift and distance modulus appropriate to the target SN.

1.5 Computational Infrastructure

All simulations are performed on the MPCDF Viper HPC cluster using SLURM job scheduling. STELLA runs on a single node with 16 GB memory and a 24 h wall-time limit. CMFGEN runs use the same resources, with dependency-chained INIT→STABLE→FINAL or single-FINAL configurations. The workflow manager `run_model_grid.py` handles automatic job cancellation of previous runs, directory cleanup, VADAT/MODEL_SPEC regeneration from YAML configuration, and consistent validation of solver parameters across all model components.

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Software STELLA (Blinnikov et al., 1998, 2006; Blinnikov & Sorokina, 2000), CMFGEN (Dessart et al., 2015; Hillier & Dessart, 2012; Hillier & Miller, 1998), KEPLER (Woosley & Heger, 2007), NumPy (Harris et al., 2020), Matplotlib (Hunter, 2007), SciPy (Virtanen et al., 2020).

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